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What Does 怀疑(*huaiyi*) Mean?: An Investigation into the Semantics of the Dubitative Verb in Mandarin Chinese

Yunchong Huang

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Abstract

This thesis attempts to investigate the semantics of the dubitative verb 怀疑(*huaiyi*) in Mandarin Chinese. Dubitative verbs have always been an interesting topic in the field of syntax and semantics cross-linguistically because of their debatable denotations and peculiar complementation patterns. 怀疑(*huaiyi*) is yet more complicated due to its ambiguity between two empirical interpretations—the first is ‘having a tendency not to believe’ (*huaiyi*₁), and the second one is ‘having a tendency to believe’ (*huaiyi*₂). Therefore, there is an additional disambiguation mechanism at work in the process of its interpretation. This thesis claims that this disambiguation mechanism is based on the information structure and is essentially related to the logical relationship between the content proposition and the immediate contextual evidence as claimed in Lu (2016), mainly depending on the location of the informational focus and the ‘givenness’ status of the content proposition in the embedding complement. As for the denotation of this verb, I claim that it is similar to the hybrid semantics of *doubt* in English proposed by Anand and Hacquard (2013), but with a crucial difference: the preference assertion is bidirectional and needs to be determined by the aforementioned disambiguation mechanism.

Key words: dubitative verb, Mandarin Chinese, disambiguation, information structure, formal semantics.

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1 Introduction: the semantic puzzles of 怀疑(huaiyi)

Dubitative predicates have always been an interesting topic in the field of syntax and semantics. The reason is that they exhibit peculiar syntactic and semantic properties cross-linguistically. Their peculiarities have been mainly discussed on two dimensions: 1) Their precise interpretations are debatable; 2) They have interesting complementation patterns. For instance, in English, there is an intuitive inclination to assume *doubt p* as the antonym of *believe p*. But when taking a closer look, it seems to have a weaker attitudinal commitment than *believe ¬p* and they are not really equivalent (Anand & Hacquard, 2013). Also, from the perspective of complementation, the English *doubt* can embed both *that*-clauses and *whether*-clauses, but disallows constituent *wh*-clauses or *whether...or not*-clauses, showing a unique pattern (Huddleston, 1994; Karttunen, 1977a, 1977b). If we switch the spotlight to the dubitative verb in Mandarin, namely the verb 怀疑(*huaiyi*), there is yet another layer of complication: it has long been observed that this verb possesses two interpretations that are nearly opposite. The first interpretation is ‘has a tendency not to believe’ (similar to *doubt* in English, marked as *huaiyi*₁), while the second one is ‘has a tendency to believe’ (similar to *suspect* in English, marked as *huaiyi*₂). Han (2001) holds that, according to native speaker intuitions, when this verb itself is shown in isolation, it is interpreted as *huaiyi*₁. Yuan (2014) claims that *huaiyi*₁ is the basic interpretation with a relatively wider distribution, while *huaiyi*₂ is derived from this basic meaning. Lu (2016), on the other hand, prefers to describe the two interpretations as composing parts of the complete semantic range of *huaiyi* triggered by different contextual information, without differentiating their status.

In sentences, this verb can embed non-interrogative complements and both of the interpretations mentioned above can be observed:

- (1) a. 他们居然怀疑这个项目的可行性。
they even *huaiyi*₁ this-CL program structural-PTCL feasibility.
‘They even doubt the feasibility of this program.’
(Yuan, 2014, p. 7)
- b. 如果有谁怀疑地心说, 那就是亵渎神灵。
if have someone *huaiyi*₁ the geocentric theory, then would be profane
god.
‘If someone doubts the geocentric theory, then it would be regarded as a profanation of the god.’
(Yuan, 2014, p. 7)
- c. 我怀疑他去了医院。
I *huaiyi*₂ he go-PERF hospital.

‘I suspect that he has gone to the hospital.’

(Feng, 2017, p. 181)

- d. 人们 怀疑 许多 不明飞行物 与 外星人 有关。
people *huaiyi*₂ many UFOs with aliens relate.
‘People suspect that many UFOs are related to aliens.’

(Yuan, 2014, p. 2)

The verb can also embed various interrogative complements, but in this research the discussion will be limited within those with ‘B-not-B’ or ‘M-not-M’ structures, in the terminology of Hagstrom (2006). Traditionally they are claimed as variants of a famous question type in Mandarin Chinese, the ‘A-not-A questions’ or ‘reduplicative questions’ named by Zhu (1994, p. 229), involving the structure of ‘two copies of a predicate with one copy negated’ (Hagstrom, 2006, p. 174), for example:

- (2) a. 小唐 喜欢 不 喜欢 小强?
Tang like NEG like Qiang?
‘Does Tang like Qiang (or not)?’
b. 小唐 喜 不 喜欢 小强?
Tang li- NEG like Qiang?
‘Does Tang like Qiang (or not)?’

The only difference between the two examples above is that (2-a) has reduplicated the whole verb predicate while (2-b) has reduplicated only the first syllable of it, which is not a word on its own. As for ‘B-not-B questions’ and ‘M-not-M questions’, the verb predicate that is copied or reduplicated in ‘A-not-A questions’ are replaced by a copula and a modal respectively. More specifically, B-not-B questions are with the structure 是不是 (*shibushi*, be+NEG+be), while M-not-M questions can take various forms based on different modals, including 会不会 (*huibuhui*, would/can+NEG+would/can) and 能不能 (*nengbuneng*, can+NEG+can). There are also ‘B- 否 (*fou*, NEG)’ and ‘M-否 (*fou*, NEG)’ structures which slightly differ from ‘B-not-B’ or ‘M-not-M’ in the semantic sense, but is preferred in written or formal expressions. Therefore, it’s not problematic to regard them as ‘quasi-B-not-B/M-not-M structures’. There are different accounts concerning the interpretation of *huaiyi* when it embeds an interrogative complement with B-not-B or M-not-M structure. This is a main reason why they are put under the spotlight in the current thesis. A group of examples from the existing literature are listed in the following, with the interpretation of *huaiyi* unsettled:

- (3) a. 副团长 对 我的 积极性
the regimental deputy commander towards I structural-PTCL enthusiasm
表示 满意, 但 仍然 怀疑 我会 不 会 割草。
express satisfaction, but still *huaiyi*? I **can NEG can** mow.
(X. Li, 1987, p. 116)
- b. 他 怀疑 自己 的 破 表 是 不 是 已经 停住,
he *huaiyi*? oneself structural-PTCL trashy watch **be NEG be** already stop,
为什么 才 11 点钟 呢?
why only 11 o'clock Q?
(X. Li, 1987, p. 116)
- c. 这 一切 老式 的 东西 让 人 怀疑 自己
these all old-fashioned structural-PTCL stuff make person *huaiyi*? oneself
是否 还 生活 在 21 世纪。
be NEG still live in 21st century.
(J. Zhang & Wu, 2013, p. 4), simplified.
- d. 人们 甚至 开始 怀疑 奥运会 能 否 按时 召开。
people even start *huaiyi*? the Olympics **can NEG** on time happen.
(Yuan, 2014, p. 4), simplified.
- e. 许多 企业 不得不 暂时 停产, 一些
many enterprises have to temporally stop producing, some
外商 怀疑 上海 经济 是否 过热。
foreign entrepreneurs *huaiyi*? Shanghai economy **be NEG** overheat.
(Yuan, 2014, p. 4), simplified.

From the illustrations above, it can be observed that there are some semantic puzzles regarding the verb *huaiyi* in Mandarin Chinese. Firstly, since there are two nearly opposite interpretations for this verb, an ambiguity arises. Given this ambiguity, when *huaiyi* embeds non-interrogative complements or interrogative complements with B-not-B/M-not-M structures, how do people choose the appropriate interpretation and comprehend the whole sentence? Is there a disambiguation mechanism at work? Secondly, if there is indeed a disambiguation mechanism, is it possible to combine this mechanism and the two empirical interpretations of the verb (*huaiyi*₁ and *huaiyi*₂) into a formal framework, so that a relatively unified semantics of *huaiyi* can be constructed? These two inquiries are the main research questions of the current thesis. The attempt to resolve these two questions will be carried out in several chapters: chapter 2 will review the existing attempts to account for the choice between the interpretations of *huaiyi*, discuss and point out their limitations. Chapter 3 will turn to related past research both in English and Chinese for a wider theoretical background. This will include a review of a compositional formal semantic research about the English dubitative *doubt* to lay basis for the later semantic construction of *huaiyi*, theories of information structure to shed light on the aforementioned potential disambiguation mechanism, and some compositional issues related to the combination of the dubitative verb and interrogative

complements. Chapter 4 will demonstrate a proposal to settle the two main research questions with insights from the chapter 2 and 3. Last but not least, the final chapter 5 will give some conclusive comments and point out some remaining issues this research could not cover.

2 Existing accounts of 怀疑(*huaiyi*)

2.1 怀疑(*huaiyi*)+non-interrogative complements

A large body of existing research has paid attention to sentences with *huaiyi* embedding a non-interrogative complement and attempted to uncover the mechanism determining the interpretation of the verb. One pioneering research is X. Li (1987), in which the author claims that the choice between the two interpretations is decided by the syntactic status of the complement embedded by the verb. When the verb embeds a nominal or pronominal complement, it is interpreted as *huaiyi*₁; whereas when the verb embeds a clausal complement, it is interpreted as *huaiyi*₂. Examples in (1) support his claim. This is so because in (1-a) and (1-b) where *huaiyi* embeds a nominal complement, the verb is interpreted as *huaiyi*₁; while in (1-c) and (1-d), with the clausal complement, *huaiyi*₂ is chosen.

However, later research has found that in cases where the verb embeds a clausal complement, the interpretation of the verb is not as uniform as X. Li (1987) claims. Han (2001) holds that sentences of this kind can be ambiguous, as two interpretations are actually both possible given different contextual information. In written texts, contextual information depends on the surrounding text; while in oral conversations, different locations of the focus are considered as contextual information making both interpretations possible, as shown in the following examples (p. 36, here the boldface represents the focus):

- (4) a. 我怀疑 明天 会 晴。
I *huaiyi*₂ tomorrow will sunny.
'I suspect that it will be sunny tomorrow.'
- b. 我怀疑 明天 会 晴。
I *huaiyi*₂ tomorrow will sunny.
'I suspect that it will be sunny tomorrow.'
- c. 我怀疑 明天 会 晴。
I *huaiyi*₁ tomorrow will sunny.
'I doubt that it will be sunny tomorrow.'

The author makes use of several means to test the examples above, concluding that when the focus is in the clausal complement (like (4-a) and (4-b)), *huaiyi* is interpreted as *huaiyi*₂; while when the focus is in the verb *huaiyi* itself (like (4-c)), it is interpreted as *huaiyi*₁. This can be regarded as a counterexample to the claim in X. Li (1987) and an empirical strategy to distinguish the two interpretations in a real use. But regrettably there is not much discussion about what the mechanism driving this phenomenon is.

Additionally, Han (2001) and Feng (2017) both claim that if *huaiyi* is modified by adverbs of degree, it is more likely to be interpreted as *huaiyi*₁. But this is actually not absolute. Park and Yuan (2016, p. 133) give examples showing that both interpretations are possible when modified by the same adverb:

- (5) a. 我很 怀疑 她 不是 谢大侠 的 女儿.
I strongly *huaiyi*₂ she NEG-be Hero Xie structural-PTCL daughter.
'I strongly suspect that she is not Hero Xie's daughter.'
- b. 我很 怀疑 一个 整天 想着 当 元帅 的
I strongly *huaiyi*₁ a-CL all day dream about being marshal structural-PTCL
士兵 会 真的 在战斗中 奋不顾身.
soldier would really in battles fight bravely.
'I strongly doubt that a soldier who dreams about being a marshal all day would fight bravely in battles.'

The fact that *huaiyi* stays ambiguous when modified by adverbs of degree is confirmed by J. Zhang and Wu (2013, p. 2) in the following example, where the authors claim that either interpretations is possible given different contexts:

- (6) 公司 失窃 了, 我们很 怀疑 是他报
company was stolen -PERF, we strongly *huaiyi*₁/*huaiyi*₂ be he report
的 案.
structural-PTCL case.
'There was a theft in the company, and we strongly doubt/suspect that it was him who reported the case.'

It is illustrated that there are two contexts suitable for the two different interpretations. In the context where *huaiyi*₁ is appropriate, after the theft, there might already be some rumours saying that 'he' was the person who reported the case, but 'we' don't agree with it. Whereas in the context where *huaiyi*₂ is appropriate, after the theft, no one knows about who reported the case, but during the investigation 'we' suspect that 'he' was the one according to some new findings at hand. Based on this and similar observations from other examples,

the authors summarize the distribution of the two interpretations based on the so-called ‘belonging relationship’ between the proposition in the complement and the subject of the sentence (the agent of *huaiyi*). It is claimed that when the proposition is put forward by the agent herself right at the point of utterance, it can be defined as ‘belonging to the agent’; while if it is put forward by others or by the agent herself in the past, then it doesn’t belong to the agent at the point of utterance. Since people would not disagree with ‘the current self’ but are able to hold different opinions regarding others or the past self, ‘belonging to the agent’ is always related to *huaiyi*₂ and ‘not belonging to the agent’ is always related to *huaiyi*₁. This can be put as follows:

- *huaiyi*₁: Subject(Agent)+*huaiyi*₁+Complement (containing a proposition made by others or the agent herself in the past)
- *huaiyi*₂: Subject(Agent)+*huaiyi*₂+Complement (containing a proposition made by the agent herself at the point of utterance)

This account seems quite attractive, but one needs to be aware that sometimes with limited contextual information, it may be difficult to determine whether the proposition in the complement ‘belongs to’ the subject (agent) at the point of uttering the sentence. Under such circumstances, the judgment about this relationship may be dependent on intuitive interpretation, causing the risk of a circular explanation. However, this rationale of assessing the informational status of the proposition is rather helpful and will be picked up again in chapter 4.

Yuan (2014) have also tried to account for the mechanism determining the distribution of two interpretations, which he claims to be originating from the effects of different complements. Although he discusses several dimensions of syntax (nominal/pronominal complements vs. clausal complements; the location of the object), semantics (degree of definiteness) and information status (new information vs. old information), he claims that the most important standard is on the dimension of pragmatics—the emotive evaluation regarding the content of the complement: when the complement is evaluated as positive or neutral, *huaiyi*₁ is chosen as the interpretation; but when the complement is evaluated as negative, *huaiyi*₂ is chosen. According to him, this originates from a tendency of ‘critical thinking developed from the pressure of evolution and social competition’ (Yuan, 2014, p. 4), which has implanted an implicit ‘build-in negative device’ in the semantics of *huaiyi*, functioning like ‘an umbrella against the sunshine of optimism’. In other words, this tendency can be summarized as ‘the tendency to believe in something bad and not to believe

in something good’ (Yuan, 2014, p. 7). He supports this claim by the following examples (p. 10, simplified):

- (7) a. 曾 有 人 怀 疑 汉 语 无 词 类
in the past have people *huaiyi*₂ Chinese language don’t have parts of speech
划分。
division.
‘There used to be people who suspect that there are no divisions of parts of speech in the Chinese language.’
- b. 曾 有 人 怀 疑 汉 语 有 词 类
in the past have people *huaiyi*₁ Chinese language have parts of speech
划分。
division.
‘There used to be people who doubt that there are divisions of parts of speech in the Chinese language.’

(7-a) is adopted from a monograph about the Chinese grammar, and in this book it’s described that there has been a traditional thinking that human languages are not ‘complete’ without divisions of parts of speech. According to this context, in (7-a) the complement ‘there are no divisions of parts of speech in the Chinese language’ is evaluated as negative, and as a result *huaiyi* in this sentence is interpreted as *huaiyi*₂. While in (7-b), the complement ‘there are divisions of parts of speech in the Chinese language’ is analyzed as positive, and *huaiyi* in this sentence is interpreted as *huaiyi*₁.

However, this criterion to distinguish the two interpretations based on pragmatic evaluations seems not to be as universal as the author claims. Counterexamples can be found as in the following:

- (8) a. 我 怀 疑 他 说 的 是 实 话。
I *huaiyi*₂ he say structural-PTCL be truth.
‘I suspect that what he said is the truth.’
- b. 我 怀 疑 他 去 了 市 场。
I *huaiyi*₂ he go ASP market.
‘I suspect that he has gone to the market.’

In (8-a), the complement embedded by *huaiyi* can be regarded as positively-evaluated, but it’s perfectly possible to interpret *huaiyi* in this sentence as *huaiyi*₂. Similarly in (8-b), the complement can be regarded as neutrally-evaluated, and *huaiyi* in this sentence can also be interpreted as *huaiyi*₂. Observations from these 2 sentences are not consistent with the claim

in Yuan (2014), indicating that it's problematic to push this 'rule of pragmatic evaluation' forward as the universal tool to distinguish the two interpretations of *huaiyi* in sentences.

Lu (2016) claims that since the dubitative attitude is essentially a logical reasoning process without full certainty, the most fundamental criterion to disambiguate the two interpretations of *huaiyi* lies in the relationship between the explicit content proposition in the embedded complement and the immediate contextual evidence (also a proposition, labelled as 'P'). If there is a reliable logical relationship such that $P \rightarrow \text{might } Q$, then when the content proposition is consistent with Q , the interpretation of *huaiyi*₂ is observed; when the following content proposition is logically $\neg Q$, *huaiyi*₁ is revealed. However, it's admitted that this criterion has its limitation in operability, since the immediate contextual evidence 'P' is not always directly accessible and the cognitive costs of this reasoning process are relatively high. Therefore, the author claims that the observations of the syntactic status and pragmatic evaluation of the complement, as X. Li (1987) and Yuan (2014) propose, are supplementary strategies. He holds that there is a hierarchy between these 3 strategies: on the axis of valid coverage and disambiguation force, the ranking is logical relationship > pragmatic evaluation > syntactic status, but from the perspective of operability, it's the other way round. It is believed that these 3 strategies are combined in the disambiguation and the interpretation of the verb.

At this stage, several existing accounts for the disambiguation mechanism of *huaiyi* in 'huaiyi+non-interrogative complement' cases are discussed, but they more or less have some limitations. Next we will turn to cases where *huaiyi* embeds B-not-B/M-not-M interrogative complements.

2.2 怀疑(*huaiyi*)+B-not-B/M-not-M interrogative complement

Some studies mentioned in the last section have also discussed 'huaiyi+B-not-B/M-not-M interrogative complement' cases with sentences in (3), although not as the main topic. In these cases, the issue becomes more complicated. Unlike in 'huaiyi+non-interrogative complement' cases, where the complement is equivalent to a proposition and the semantics of *huaiyi* combines with the proposition directly, interrogative complements are regarded as questions, and a classic semantic theory of questions put forward by Hamblin (1958, 1973) claims that they are sets of propositions. Following this, a question with an 'A-not-A', 'B-not-B' or 'M-not-M' structure (either prototypical or quasi) can be regarded as a proposition set containing one proposition and its negation, since this type of questions in Mandarin are most naturally answered by repeating either the questioned predicate/copula/modal or

its negation (Hagstrom, 2006) and ‘a question picks out a set of propositions that count as answers’ (Hagstrom, 2003, p. 188). Therefore, in this case, before discussing the unresolved disambiguation mechanism of *huaiyi*, one should work out what is combined with *huaiyi* in the first place. Is *huaiyi* combined with the set of propositions represented by the interrogative complement as a whole, or with one certain proposition from the set?

When discussing the sentences in (3), the existing literature haven’t made overt claims regarding this compositional problem. Instead, they directly dive into the interpretation problem of *huaiyi*. A review of these discussions will be given, and some implicit claims about the compositional problem can be observed in the process.

J. Zhang and Wu (2013) claim that there is a semantic continuum of *huaiyi*: ‘*huaiyi*₁ (doubt)-wonder-*huaiyi*₂ (suspect)’. They hold that the neutralized intermediate interpretation of ‘wonder’ is triggered when *huaiyi* embeds an interrogative complement with the prototypical or quasi ‘B-not-B’ or ‘M-not-M’ structure, conveying uncertainty between the affirmative proposition and its negation. This claim essentially holds that *huaiyi* in this case combines with the set of propositions as a whole. (3-c) is the example used by the authors to illustrate their claim. The idea is that *huaiyi* in this sentence conveys neutral uncertainty. However, the existence of this neutral interpretation is actually debatable. Although it does seem acceptable to translate (3-c) as ‘All these old-fashioned stuff make one wonders whether she still lives in the 21st century or not’, a more intuitive reading for this sentence exists: since 老式 (‘old-fashioned’) in the sentence gives a contextual clue, it can be safely inferred that the agent somehow tends to doubt (*huaiyi*₂), at least for a moment, that she is living in the 21st century, given the environment with a lot of old-fashioned stuff around. This intuitive reading is against the account of a neutral stand, revealing that the verb is still biased to *huaiyi*₂ and is combined with one certain proposition from the set.

The intuitive reading is supported by Yuan (2014). He claims that, in real communications, the interpretation of *huaiyi* always leans to either *huaiyi*₁ or *huaiyi*₂ and they are eventually combined with a specific proposition, leaving no space for a ‘wonder’ interpretation expressing neutral uncertainty. Extending the rationale of regarding the pragmatic evaluation of the complement as the important criterion for interpretation, he claims that this standard can also explain this phenomenon: when *p*, standing for the affirmative proposition in the set represented by the complement, is positively or neutrally evaluated, the interpretation tends to be ‘*huaiyi*₁ + *p*/*huaiyi*₂ + $\neg p$ ’; while when *p* is negatively evaluated, the interpretation tends to be ‘*huaiyi*₂ + *p*/*huaiyi*₁ + $\neg p$ ’. Applying this to (3-d) and (3-e) produces interpretations that are consistent with the intuitions of native speakers: in (3-d), the *p* of the complement 奥运会能否按时召开 (‘whether the Olympics can happen on time’) is

奥运会能按时召开 (‘the Olympics can happen on time’), while the $\neg p$ is 奥运会不能按时召开 (‘the Olympics cannot happen on time’); as p here is positively evaluated, the matrix VP is interpreted as ‘ $huaiyi_1 + p/huaiyi_2 + \neg p$ ’; in (3-e) the p of the complement 上海经济是否过热 (‘whether the economy of Shanghai is overheated’) is 上海经济过热 (‘the economy of Shanghai is overheated’), while the $\neg p$ is 上海经济没有过热 (‘the economy of Shanghai is not overheated’), as p here is negatively evaluated, the matrix VP is interpreted as ‘ $huaiyi_2 + p/huaiyi_1 + \neg p$ ’.

There is a somewhat similar account given by X. Li (1987) for ‘*huaiyi*+interrogative complement’ situations, using the notion of ‘desirability’. He claims that interrogative complements with ‘B-not-B’ or ‘M-not-M’ structures are essentially euphemisms to convey some ‘undesirable’ declarative message with the affirmative proposition or its negation within the sets they denote. For example, in (3-a) the p of the complement 我会不会割草 (‘whether I can mow’) is 我会割草 (‘I can mow’), while the $\neg p$ is 我不会割草 (‘I can’t mow’). Here $\neg p$ is regarded as something ‘undesirable’, since the overall declarative message should be ‘undesirable’ in need of a euphemism, the interpretation should be ‘ $huaiyi_2 + \neg p$ ’, meaning that the sentence is actually interpreted as ‘The regimental deputy commander..., but still suspected that I can’t mow’. In (3-b), on the other hand, the p of the complement 自己的破表是不是已经停住 (‘whether his trashy watch has stopped’) is 自己的破表已经停住 (‘his trashy watch has stopped’), while the $\neg p$ is 自己的破表没有停住 (‘his trashy watch has not stopped’). Here p is regarded as something ‘undesirable’, and with the same rationale that has been mentioned, the interpretation should be ‘ $huaiyi_2 + p$ ’, meaning that the sentence is actually interpreted as ‘He suspected that his trashy watch has stopped, or else why was it only 11 o’clock?’.

However, just as what has been observed in ‘*huaiyi*+non-interrogative complement’, the accounts proposed by Yuan (2014) and X. Li (1987) face empirical problems. For example, they fail to account for the following sentences:

- (9) a. 小黄 怀疑 小唐 是 不 是 喜欢 小强。
Huang *huaiyi*? Tang be NEG be like Qiang.
b. 小唐 怀疑 小邱 会 不 会 弹 吉他。
Tang *huaiyi*? Qiu can/would NEG can/would play the guitar.

It’s observed that these two examples above can have both the ‘ $huaiyi_1 + p/huaiyi_2 + \neg p$ ’ and ‘ $huaiyi_2 + p/huaiyi_1 + \neg p$ ’ interpretations. In oral utterance, when the phonetic stress is on 是不是 (be-NEG-be) or 会不会 (can/would-NEG-can/would), the former interpretation is preferred; while when the phonetic stress is on other elements of the complement, the

latter is preferred. Additionally, the evaluation of the proposition, either ‘positive/negative’ or ‘desirable/undesirable’ can be quite subjective due to the lack of more contextual information. For instance in (9-a), 小唐喜欢小强 (p , ‘Tang likes Qiang’) or 小唐不喜欢小强 ($\neg p$, ‘Tang doesn’t like Qiang’) are possible to have different evaluations for different speakers in various contexts. As a result, a more general explanation is worth investigating, and the observation that different phonetic emphasises can alter the interpretation of a sentence in the same surface form, resonating with the observation by Han (2001) in (4), may be an important clue.

Since the accounts above given by the two authors are not unproblematic, their assumption ‘*huaiyi* picks out and combines with one proposition from the set represented by the B-not-B/M-not-M interrogative complement’ also needs to be reviewed to see whether it’s justifiable. This will be discussed in the following chapter.

3 Theoretical background

3.1 The semantics of dubitative predicates in English

By now we have mainly discussed the disambiguation puzzle of *huaiyi*, which largely concerns the first main research question. However, in order to construct a relatively comprehensive formal semantic account for this verb, one needs to settle a more basic problem: how to accommodate the two empirical interpretations of the verb into a formal framework? Answering this is essential for the solution of the second research question.

huaiyi is a dubitative predicate. As have mentioned, empirical evidence show that it denotes attitude relations that would be translated as *doubt* and *suspect* in English in a same lexical form. Therefore, existing research for English dubitatives will be useful for our purposes. Anand and Hacquard (2013) is a good reference as such. The article first reviews a classic semantic split in attitude verbs, namely the division between ‘representational’ and ‘non-representational’ ones (Bolinger, 1968). The former category refers to those that convey a mental picture and quantify over an information state, such as *believe* and *know*; while the latter category includes those verbs that combine with their complement through a comparison of preference with alternatives provided by the context without providing an information state, such as *want*. On the basis of this, the authors further claim that emotive doxastics (such as *hope* and *fear*) and dubitatives (such as *doubt*) have hybrid semantics with traits of both the categories aforementioned: they do quantify over information states, but in an existential rather than a universal way; meanwhile they also have a preference

component, as emotive doxastics contain emotive preferences and dubitatives are with preference of probability. More specifically, they offer a formal semantic account for *doubt* when embedding a declarative complement as follows (p. 36):

- (10) $\llbracket \text{a doubts}_C \text{ that } \varphi \rrbracket^{c,w,S,g}$ is defined iff
 φ -verifiers in $S' \neq \emptyset \wedge \varphi$ -falsifiers in $S' \neq \emptyset$. [Uncertainty Condition]
 If defined = 1 iff
 $\exists w' \in S' : [\llbracket \varphi \rrbracket^{c,w',S,g} = 1] \wedge$ [Doxastic Assertion]
 p -falsifiers $>_{PROB_{a,w}}$ p -verifiers [Preference Assertion]
 where $S' = DOX_{a,w}$ and φ -verifiers in $S' =$
 $\lambda S'' . S'' \subset S' \wedge \forall S''' \subset S'' : [\forall w' \in S''' : [\llbracket p \rrbracket^{c,w',S''',g} = 1]]$ ($= \text{pow}(S' \cap p)$)
 φ -falsifiers in $S' = \neg \varphi$ -verifiers in S'

As can be seen, this account possesses an assertion of doxastic possibility, represented by an existential quantifier instead of a universal quantifier over possible worlds, requiring only some of them to verify p , namely the explicit content proposition in the complement. In addition, to secure this uncertainty in the doxastic state, neither the worlds verifying the complement containing p (φ -verifiers) nor the worlds falsifying it (φ -falsifiers) in the subject's doxastic state should be trivial, which is named as 'the uncertainty condition'. Last but not least, when interpreting *doubt*, there is clearly a tendency for the subject to take the p -falsifiers as more probable to hold than the p -verifiers, which is named 'the preference condition'. This account is *prima facie* applicable for the interpretation of *huaiyi*, since among the two interpretations of it, *huaiyi*₁ is similar to *doubt*, and *huaiyi*₂ is similar to *suspect*, which can also be explained in a similar manner, possibly with a reverse preference assertion.

3.2 Related concepts of information structure

It has been observed that the attempts above to account for the disambiguation mechanism of the two interpretations of *huaiyi* all have some limitations. However, when taking a closer look, there are traits discussed by these attempts that can be included in a common theoretical framework. Han (2001) mentions the influence on the interpretation brought by different locations of the prosodic focus; J. Zhang and Wu (2013) explains the important role that the contextual information plays in determining the interpretation; in Yuan (2014), although the pragmatic evaluation is regarded as the most important criterion, dimensions of information status and degree of definiteness are also mentioned. Lu (2016) have also mentioned the

relationship between *huaiyi*₁ and old knowledge and the relationship between *huaiyi*₂ and new information. All these can be discussed within the framework of information structure.

In the theory of information structure, the ‘informational focus’ is an important notion. Usually in a sentence, the part labeled as the informational focus carries information which is new or prominent, and a classic method to identify a potential informational focus is to see whether it can correspond to the part that answers a wh-constituent constituent question (Halliday, 1967). In Mandarin Chinese, the informational focus is basically expressed with phonetic stress while leaving the focused element in-situ (Badan, 2015). More specifically, phonological research have found that elements within the informational focus are with enlarged pitch range and longer duration (K. Li, 2009; H. Zhang & Pan, 2019). Additionally, based on the notion of informational focus, the concept of ‘focus semantic value’ is developed in the theory of alternative semantics. Rooth (1992, 2016, 1985) assumes that the focus marking introduces a contextually-restricted set of alternatives to the focused constituent, which is the so-called ‘focus semantic value’, with the original denotation of this constituent (named as the ‘ordinary semantic value’) included. To model the composition of focus semantic values, a recursive strategy of ‘alternatives all the way down’ is proposed, where non-focused phrases have their focus semantic value of a singleton set containing their ordinary semantic value. The following example is used for illustration, with the subscript F standing for the focus, the superscript F standing for the focus semantic value and the superscript O standing for the ordinary semantic value (adopted from Hinterwimmer (2012, p. 1878), with alternations):

- (11) a. $[[\text{married Bertha}]]^O = \lambda x. \lambda w. \text{married} (x, \text{Bertha}) (w)$
 b. $[[[_F \text{married Bertha}]]^F = \{ \lambda x. \lambda w. \text{married} (x, \text{Bertha}) (w), \lambda x. \lambda w. \text{won} (x, \text{Nobel Prize}) (w), \lambda x. \lambda w. \text{sang} (x) (w), \dots \}$
 c. $[[\text{married Bertha}]]^F = \{ \lambda x. \lambda w. \text{married} (x, \text{Bertha}) (w) \}$
 d. $[[\text{John } [_F \text{married Bertha}]]^O = \lambda w. \text{married} (\text{John}, \text{Bertha}) (w)$
 e. $[[\text{John } [_F \text{married Bertha}]]^F = \{ \lambda w. \text{married} (\text{John}, \text{Bertha}) (w), \lambda w. \text{won} (\text{John}, \text{Nobel Prize}) (w), \lambda w. \text{sang} (\text{John}) (w), \dots \}$

Another important concept is ‘givenness’, basically a property indicating that the denotation of a constituent has either been introduced explicitly or implicitly (in the sense of its existence being entailed by the common ground) into the discourse. This property can be revealed in several means. A common one of them is phonetic de-accentuation, while others include pronominalizations and definite references of discursive elements on the grammatical dimension (Hinterwimmer, 2012; Schwarzschild, 1999).

Potentially, the theory and concepts of information structure illustrated in this section can be combined with the empirical observations from the literature to uncover the disambiguation mechanism of *huaiyi*, to which we will turn in the chapter 4.

3.3 Compositional issues of interrogative complements

In section 2.2 it has been mentioned that there is an additional compositional uncertainty when *huaiyi* embeds interrogative complements with B-not-B/M-not-M structures. Following the traits of these interrogatives introduced in 2.2, similar types of embedded interrogative complements can be found in English, namely the *whether ...* or *whether ... or not* structure. One thing worth noting is that embedded interrogative complements can be regarded as non-root questions. Following Hamblin semantics, it's assumed that the prejacent of the question-operator has a proposition-set denotation (Hamblin, 1973; Rooth, 1985).

There have been interesting observations about the cases where *whether ...* and *whether ... or not* structures are combined with different verb predicates, for instance, *doubt* and *know* in the following examples:

- (12) a. Ann doubts that it is raining.
b. Ann doubts whether it is raining.
c. *Ann doubts whether it is raining or not.
d. *Ann doubts what the weather is.
- (13) a. Ann knows that it is raining.
b. Ann knows whether it is raining.
c. Ann knows whether it is raining or not.
d. Ann knows what the weather is.

The grammaticality patterns above require an explanation, and speculations have been made to claim that the differences originate from different natures of predicates and embedded interrogative complements. Karttunen (1977a) holds that *wh*-questions and alternative questions are two main kinds of questions, while 'yes/no' questions are regarded as 'degenerated' alternative questions. He observes that the two main kinds of questions are with the same syntactic distribution, since verbs in general have a conformity in the grammaticality of sentences where they embed *wh*-questions and *whether*-questions (as 'yes/no' questions, the 'degenerated' alternative questions) respectively, and those that can embed both are regarded

as ‘question-embedding verbs’ (like *know*). However, *doubt*, along with other dubitative verbs, is an exception to this as they can embed *whether*-questions (like (12-b)) but not *wh*-questions (like (12-d))¹. Later in Karttunen (1977b) he further observes that unlike ‘question-embedding verbs’ like *know*, disjunctive *whether*-questions (*whether...or...*) and ‘extended’ *whether*-questions (*whether...or not*) cannot be embedded by *doubt*, and *whether* in sentences like (12-b) can be perfectly replaced by *that* without altering the meaning. In addition, *doubt* is combined with this content proposition rather than with whichever proposition that is true in the evaluation world from the set represented by the *whether*-question, as in cases of ‘question-embedding verbs’. As a conclusion, when embedded by *doubt*, *whether*-complements are not taken as genuine questions and *doubt* is regarded to be outside the category of ‘question-embedding verbs’.

Biezma and Rawlins (2012) have inherited the distinction between *doubt* and other ‘question-embedding verbs’, but have provided more refined categories of question types and a uniform analysis of *whether*-complements. Based on Biezma (2009), *whether ...* structures are analyzed as embedded polar questions ($P_{OL}Qs$), while *whether...or not* structures are regarded as a special type of embedded alternative questions formed by two opposite alternatives p and $\neg p$ ($A_{LT}Q_{VNs}$). The claim is that they are with different question operators implementing different semantic treatments and presuppositions regarding the ‘salient alternative sets’, as shown below (Biezma & Rawlins, 2012, pp. 388, 392–393):

- (14) **Definition:** $\text{SalientAlts}(c)$ is the set of propositional alternatives that are salient in the context of interpretation c .

Constraint on salient propositional alternative sets in a context c :

- (i) $\forall w \in cs_c : \exists p \in \text{SalientAlts}(c) : p(w) = 1$ [Exhaustivity]
(ii) $\forall w \in cs_c : \forall p, q \in \text{SalientAlts}(c) : (p = q \vee \neg(p(w) \wedge q(w)))$ [Mutual Exclusivity]

- (15) a. **Polar Question Operator:**

$$\llbracket [I_{[Q]} \alpha] \rrbracket^c = \llbracket \alpha \rrbracket^c$$

defined only if

- (i) $\llbracket \alpha \rrbracket^c \subseteq \text{SalientAlts}(c)$ or if $\text{SalientAlts}(c) = \emptyset$, and

- (ii) $|\llbracket \alpha \rrbracket^c \cup \text{SalientAlts}(c)| > 1$

- b. **Closure Operator for Alternative Questions:**

$$\llbracket [I_{[Q]} \alpha]_{H*L-L} \rrbracket^c = \llbracket [I_{[Q]} \alpha] \rrbracket^c$$

defined only if $\text{SalientAlts}(c) = \llbracket [I_{[Q]} \alpha] \rrbracket^c$

¹Karttunen (1977a) also mentions that ‘emotive factives’ in English, such as *be amazing*, *be surprising* and *bother* don’t conform to the aforementioned pattern. They can embed *wh*-questions but not alternative questions.

(Presupposition: listed alternatives have exhausted all salient alternatives)

Constraint: α must contain a disjunction.

As can be seen in (15-a), although admitting their status as ‘questions’, *POLQs* (including *whether ...* structures) are specially treated as singleton sets, mainly out of two reasons: (i) they compositionally lack any alternative-introducing item, for instance disjunctions or other devices of disassociation; (ii) the higher utility value of (or the speaker’s preference for) the content proposition than its negation (van Rooy & Safárová, 2003). As a result, embedding *POLQs* are equivalent to non-interrogative complements, being singleton sets in the sense of alternative semantics. This explains the reason why *doubt*, not being a ‘questions-embedding’ verb, can take an embedded *POLQ* as its complement as in (12-b).

However, since whether a verb is ‘questions-embedding’ or not cannot be predicted solely on the basis of their denotations, one may perceive that the accounts above are rather stipulative. In contrast, Pruitt and Roelofsen (2011) take another approach, assuming that all verbs are question-embedding and *POLQs* are still sets with two members. But differences can still be observed between sentences with *doubt* and those with *know*. Take the following as examples:

- (16) a. John doubts whether the door is open.
 b. John doubts whether the door is closed.
 c. *John doubts whether the door is open or closed.
 d. *John doubts whether the door is open or not.
- (17) a. John knows whether the door is open.
 b. John knows whether the door is closed.
 c. John knows whether the door is open or closed.
 d. John knows whether the door is open or not.

It’s observed that (16-a) and (16-b) are not truth-conditionally equivalent, since ‘in a situation where John suspects that the door is open, we can truthfully say that he doubts whether the door is closed, but not that he doubts whether the door is open’ (Pruitt & Roelofsen, 2011, p. 27). This can’t be explained if $\llbracket$ whether the door is open $\rrbracket$ and $\llbracket$ whether the door is closed $\rrbracket$ stand for the same proposition set according to the traditional Hamblin account. Therefore, a ‘highlighting’ mechanism is introduced: though embedded *POLQs* $\llbracket$ whether the door is open $\rrbracket$ and $\llbracket$ whether the door is closed $\rrbracket$ essentially stand for the same proposition set, the one and only highlighted proposition in them are different according to their surface form.

Coincidentally, from the semantic perspective, the predicate *doubt* has a constraint in its denotation that it must combine with the one and only highlighted proposition from the set when it embeds an interrogative complement, causing the non-equivalence between (16-a) and (16-b). Meanwhile, since embedded alternative questions and $A_{LT}Q_VNs$ highlight both alternatives in the set (as revealed in the surface form), (16-c) and (16-d) are ungrammatical due to the conflict between their highlighting mechanism and the denotation of *doubt*.

On the other hand, (17-a) and (17-b) are truth-conditionally equivalent, since ‘John knows whether the door is open if and only if he knows whether the door is closed’ (Pruitt & Roelofsen, 2011, p. 27). This can be attributed to the absence of the aforementioned constraint proposed for *doubt* here in the denotation of *know*, so that this verb can embed any interrogative complement without further requirements for their specific highlighting. This also explains the grammaticality of (17-c) and (17-d).

To summarize, the idea of Pruitt and Roelofsen (2011) is that the highlighting mechanism of interrogative complements is always there, while the results of composing interrogative complements with a verb are influenced by the details of the verb’s denotation, for example, whether the distribution of ‘highlights’ is relevant. This account can be helpful for a proposal concerning the composition of *huaiyi* and interrogative complements in the later chapter.

4 Proposals for the semantics of 怀疑(*huaiyi*)

4.1 Disambiguation mechanism based on information structure

4.1.1 The mechanism in ‘怀疑 (*huaiyi*)+non-interrogative complement’

With the theoretical background of information structure in the last chapter, we are now prepared to re-examine some example sentences listed in previous chapters to approach the potential disambiguation mechanism. As have mentioned, phonetic stress is an important cue for informational focus, while phonetic de-accentuation conveys ‘givenness’. We have seen in Section 2.1 the claim by J. Zhang and Wu (2013), which identifies the interpretation of *huaiyi* according to the ‘belonging relationship’ between the proposition and the agent at the point of utterance. Although it has been stated that this ‘belonging relationship’ concept can be subjective, the rationale of assessing the proposition based on its informational status in the contexts can be utilized here. It turns out that among cases where *huaiyi* embeds non-interrogative complements, *huaiyi*₁ is always chosen when the informational focus is not in the proposition and the proposition is marked as ‘given’, whereas when *huaiyi*₂ is chosen,

the informational focus is within the proposition and the proposition is marked as ‘not given’. This can be supported by the reanalysis of examples in (4) offered by Han (2001):

- (18) a. 我怀疑 $[_F\text{明天 会 晴}]$ 。
 I *huaiyi*₂ tomorrow will sunny.
 ‘I suspect that it will be sunny tomorrow.’
The ‘givenness’ status of the proposition: not given
 $[[\text{怀疑 } [_F\text{明天会晴}]]^F = \{\lambda w \lambda x. x \text{ } huaiyi \text{ that it will be sunny tomorrow in } w, \lambda w \lambda x. x \text{ } huaiyi \text{ that Hong can't drink in } w, \lambda w \lambda x. x \text{ } huaiyi \text{ that Ming has left in } w, \dots\}$
 $[[[_F\text{明天会晴}]]^F = \{\lambda w. \text{ It will be sunny tomorrow in } w, \lambda w. \text{ Hong can't drink in } w, \lambda w. \text{ Ming has left in } w, \dots\}$
 $\Rightarrow | [[[_F\text{明天会晴}]]^F | > 1$
The chosen interpretation: *huaiyi*₂
- b. 我 $[_F\text{怀疑}]$ 明天 会 晴。
 ‘I doubt (*huaiyi*₁) that it will be sunny tomorrow.’
The ‘givenness’ status of the proposition: given
 $[[[_F\text{怀疑}] \text{明天会晴}]]^F = \{\lambda w \lambda x. x \text{ } huaiyi \text{ that it will be sunny tomorrow in } w, \lambda w \lambda x. x \text{ } \textit{‘believes’} \text{ that it will be sunny tomorrow in } w, \lambda w \lambda x. x \text{ } \textit{‘predicts’} \text{ that it will be sunny tomorrow in } w, \dots\}$
 $[[\text{明天会晴}]]^F = \{\lambda w. \text{ It will be sunny tomorrow in } w\}$
 $\Rightarrow | [[\text{明天会晴}]]^F | = 1$
The chosen interpretation: *huaiyi*₁

Therefore, it can be observed that when *huaiyi*₂ is chosen, the phonetic stress is on the proposition. This brings the analysis that the informational focus lies in the proposition, and since the proposition is accentuated, it should be regarded as ‘not given’. As a result, the focus semantic value of the matrix VP is a set in which the alternatives of the object of *huaiyi* are introduced, and the focus semantic value of the embedded CP is a set with plural members. On the other hand, when *huaiyi*₁ is chosen, the picture is the other way round: the phonetic stress is on the predicate *huaiyi* while the proposition is de-accentuated, so the informational focus is analyzed to be on *huaiyi* and the proposition is marked as ‘given’. Accordingly, the focus semantic value of the matrix VP is a set with the alternatives of the predicate introduced, while the focus semantic value of the CP is a trivial singleton set since it’s not focused.

A relationship between the distinctions in information structure analyzed above and the interpretation selection of *huaiyi* seems to be in place, but more empirical evidence is needed to make this relationship reliable. Since sentences in 2.1 are used to support existing accounts and to point out their limitations, if this relationship can be observed in all of them, then it would be promising to reveal the disambiguation mechanism when interpreting *huaiyi* on its basis. We now turn to reanalyze the sentence which J. Zhang and Wu (2013) claim to be ambiguous. It turns out that the phonetic stress does play a role in the disambiguation, and it's consistent with contextual factors the authors claim, which is illustrated in 2.1:

- (19) a. 公司 失窃 了, 我们很 怀疑 是 [_F他] 报
company was stolen -PERF, we strongly *huaiyi*₂ be he report
的 案。
structural-PTCL case.
'There was a theft in the company, and we strongly suspect that it was him who reported the case.'

The 'givenness' status of the proposition: not given

$\llbracket \text{很 怀疑 是 } [\text{他}] \text{ 报 的 案} \rrbracket^F = \{\lambda w \lambda x. x \text{ strongly } \textit{huaiyi} \text{ that it was him who reported the case in } w, \lambda w \lambda x. x \text{ strongly } \textit{huaiyi} \text{ that it was the guard who reported the case in } w, \lambda w \lambda x. x \text{ strongly } \textit{huaiyi} \text{ that it was Ming who reported the case in } w, \dots\}$

$\llbracket \text{是 } [\text{他}] \text{ 报 的 案} \rrbracket^F = \{\lambda w. \text{ It was him who reported the case in } w, \lambda w. \text{ It was the guard who reported the case in } w, \lambda w. \text{ It was Ming who reported the case in } w, \dots\}$

$\Rightarrow |\llbracket \text{是 } [\text{他}] \text{ 报 的 案} \rrbracket^F| > 1$

The chosen interpretation: *huaiyi*₂

- b. 公司失窃了, 我们很 [_F怀疑] 是他报的案。
'There was a theft in the company, and we strongly doubt (*huaiyi*₁) that it was him who reported the case.'

The 'givenness' status of the proposition: given

$\llbracket \text{很 } [\text{怀疑}] \text{ 是 他 报 的 案} \rrbracket^F = \{\lambda w \lambda x. x \text{ strongly } \textit{huaiyi} \text{ that it was him who reported the case in } w, \lambda w \lambda x. x \text{ strongly 'believes' that it was him who reported the case in } w, \lambda w \lambda x. x \text{ strongly 'hopes' it was him who reported the case in } w, \dots\}$

$\llbracket \text{是他报的案} \rrbracket^F = \{\lambda w. \text{ It was him who reported the case in } w\}$

$\Rightarrow |\llbracket \text{是他报的案} \rrbracket^F| = 1$

The chosen interpretation: *huaiyi*₁

Observations above confirm the potential relationship with a further clarification in details: in cases where *huaiyi*₂ is chosen, it's not necessary for the whole proposition to be the informational focus. Instead, what really matters is for the informational focus marked by the phonetic stress to be within the proposition. When this is the case, the proposition would by no means be 'given' since at least some parts of it are new and prominent information, being accentuated. This also coincides with the authors' claim that it needs to be new to the context ('no one had claimed to know who reported the case'). Eventually, this leads to the fact that the focus semantic value of the proposition is a set with multiple members. For *huaiyi*₁ cases, it's largely the other way round: with no informational focus within it, the proposition can be marked as given and something already known in the context, and the result is that its focus semantic value is a trivial singleton set.²

As for examples in (7) that Yuan (2014) uses to support his claim of pragmatic evaluation and counterexamples in (8), they all actually have two possible interpretations and the relationship stated above can also be observed:

- (20) a. 曾 有 人 怀 疑 汉 语 [F无 词类
in the past have people *huaiyi*₂ Chinese language don't have parts of speech
划分]。
division.
'There used to be people who suspect that there are no divisions of parts of
speech in the Chinese language.'
- b. 曾 有 人 [F怀疑] 汉 语 无 词 类 划 分。
'There used to be people who doubt (*huaiyi*₁) that there are no divisions of
parts of speech in the Chinese language.'
- (21) a. 曾 有 人 怀 疑 汉 语 [F有 词类
in the past have people *huaiyi*₂ Chinese language have parts of speech
划分]。
division.
'There used to be people who suspect that there are divisions of parts of speech
in the Chinese language.'
- b. 曾 有 人 [F怀疑] 汉 语 有 词 类 划 分。
'There used to be people who doubt (*huaiyi*₁) that there are divisions of parts
of speech in the Chinese language.'

²One may wonder if it's possible for the informational focus to be neither on *huaiyi* nor within its complement. Empirically, it's necessary for at least one of them to be the informational focus, and the distribution is complementary. But extra informational focus other than the ones on them are possible.

- (22) a. 我怀疑 他说的 $[_F \text{是实话}]$ 。
 I $huaiyi_2$ he say structural-PTCL be truth.
 ‘I suspect that what he said is the truth.’
 b. 我 $[_F \text{怀疑}]$ 他说的是实话。
 ‘I doubt ($huaiyi_1$) that what he said is the truth.’
- (23) a. 我怀疑 他去了 $[_F \text{市场}]$ 。
 I $huaiyi_2$ he go ASP market.
 ‘I suspect that he has gone to the market.’
 b. 我 $[_F \text{怀疑}]$ 他去了市场。
 ‘I doubt ($huaiyi_1$) that he has gone to the market.’

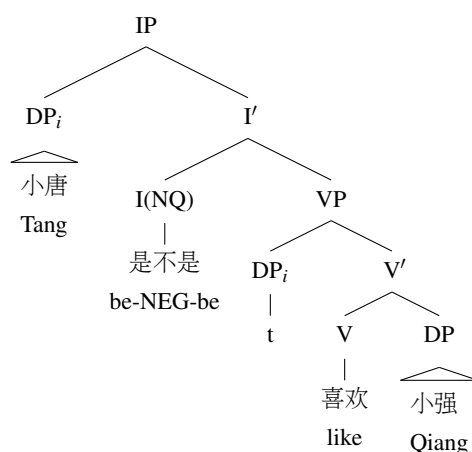
To summarize, on the formal semantic dimension, resulted by the different distributions of informational focus and the different markings of the proposition’s ‘givenness’, the mechanism disambiguating the two interpretations of *huaiyi* in ‘*huaiyi*+non-interrogative complement’ cases lies in the number of members in the set denoted by the proposition’s (equivalent to CP in this case) focus semantic value: when $|\llbracket \text{CP} \rrbracket^F| > 1$, $huaiyi_2$ is chosen; while when $|\llbracket \text{CP} \rrbracket^F| = 1$, $huaiyi_1$ is chosen.

4.1.2 The mechanism in ‘怀疑 (*huaiyi*)+B-not-B/M-not-M interrogative complement’

In Section 3.3 we have gone through some compositional insights from cases with *doubt* embedding *whether...* structure. However, to apply these insights to our purpose, a similarity between ‘*huaiyi*+B-not-B/M-not-M interrogative complement’ cases in Mandarin and English cases we have discussed must be reached. We claim that B-not-B/M-not-M interrogative complements in Mandarin are similar to *whether...* complements in English.

Firstly, they are similar in the logical form. B-not-B/M-not-M structures in Mandarin can also be treated as question operators scoping over propositions, just like *whether*. Additionally, Huang (1991) claims that A-not-A sequences, so does B-not-B/M-not-M structures, are morphological exponents that should be regarded as ‘interrogative inflections’ (labeled as ‘NQ’, reflecting that it involves both negative and interrogative characters). On the basis of this, taking (9-a) as an example, its B-not-B complement can be analyzed as follows:

(24)



Notice that 小唐(*xiaotang*), the subject of the proposition, is pronounced at the higher location through a movement from its original location in the VP, where a trace is left behind. While on the dimension of logical form it still resides in the VP. Therefore, the complement can essentially be written as $\llbracket \text{是不是}_Q[\text{小唐}_i \text{ 喜欢 小强}] \rrbracket$, similar to *whether...* complements in which *whether* scopes over a proposition as a question operator.

Secondly, when examining the ‘NQ’ operator closer, McCawley (1994) claims that the negative morpheme in its reduplicative form is real in the sense of morphology, since its specific form is influenced by aspectual structure and lexical restrictions of the predicate. However, Hagstrom (2006, p. 179) claims that it ‘does not decisively present in syntax and interpretation’, coinciding with (24) where the ‘NQ’ as a whole is treated as an operator. One supporting evidence for this claim lies in the ‘cornering effect’ proposed by Biezma (2009) as an important trait that can differentiate $A_{LT}Q_VNs$ from $P_{OL}Qs$ — this effect is not obvious in Mandarin B-not-B/M-not-M interrogatives, since they don’t seem to enforce an obligatory constraint for the other party to answer within the alternatives provided, and conversational strategies to respond indirectly are possible. Therefore, it can be assumed that B-not-B/M-not-M structures in Mandarin are more similar to *whether...* complements (embedded $P_{OL}Qs$) rather than *whether...or not* complements (embedded $A_{LT}Q_VNs$).³

With this similarity reached, along with the fact that *huaiyi* is also a dubitative verb, it’s then possible to introduce the highlighting mechanism proposed by Pruitt and Roelofsen (2011) to account for the composition in Mandarin ‘*huaiyi*+B-not-B/M-not-M interrogative

³ Another supporting evidence can be found in Schaffer and Chen (2001, p. 840), in which it is claimed that when uttering a B-not-B question, the speaker has a ‘subjective suspicion’ leaning towards the following content proposition. This coincides with the important criterion of ‘imbalanced utility value’ of the proposition and its negation for English $P_{OL}Qs$ proposed by van Rooy and Safárová (2003). But little research has discussed M-not-M questions from this perspective.

complement' cases. Here I hypothesise that a B-not-B/M-not-M interrogative complement denotes a set containing the content proposition following the NQ-operator and its negation, but with only the content proposition highlighted.

Having made progress in the compositional issue, the disambiguation mechanism of *huaiyi* when embedding B-not-B/M-not-M interrogative complement can now be considered. Similar to the last section, we will also try to pursue an account based on the information structure. This will also be done by reanalyzing related examples from different sources discussed above to guarantee the coverage. It turns out that the status of the content proposition regarding the informational focus and 'givenness' also plays an important role in the disambiguation, forming a parallel with '*huaiyi*+non-interrogative complement' cases. Let's first start by '*huaiyi*+prototypical B-not-B interrogative complement' with (3-b) and (9-a), repeated here:

- (25) a. 他怀疑 自己的 破表 是不是已经
 he *huaiyi*₂ oneself structural-PTCL trashy watch be NEG be already
 [_F停住], 为什么才 11 点钟 呢?
 stop, why only 11 o'clock Q?
 'He suspected that his trashy watch has stopped, or else why was it only 11 o'clock?'
The 'givenness' status of the proposition: not given
 $\llbracket \text{是不是}_Q [\text{破表 已经 } [\sub{F}\text{停住}]] \rrbracket^F$
 $= \{ \{ \lambda w. \text{The trashy watch has stopped in } w, \lambda w. \text{The trashy watch has not stopped in } w \}, \{ \lambda w. \text{The trashy watch was broken in } w, \lambda w. \text{The trashy watch was not broken in } w \}, \dots \}$
 $\Rightarrow | \llbracket \text{是不是}_Q [\text{破表 已经 } [\sub{F}\text{停住}]] \rrbracket^F | > 1$
The chosen interpretation: *huaiyi*₂
- b. 他怀疑 自己的 破表 [_F是不是] 已经停住。
 'He doubted (*huaiyi*₁) that his trashy watch has stopped.'
The 'givenness' status of the proposition: given
 $\llbracket [\sub{F}\text{是不是}_Q] [\text{破表 已经 停住}] \rrbracket^F$
 $= \{ \{ \lambda w. \text{The trashy watch has stopped in } w, \lambda w. \text{The trashy watch has not stopped in } w \} \}$
 $\Rightarrow | \llbracket [\sub{F}\text{是不是}_Q] [\text{破表 已经 停住}] \rrbracket^F | = 1$
The chosen interpretation: *huaiyi*₁
- c. 小黄 怀疑 小唐 是不是 [_F喜欢] 小强。
 Huang *huaiyi*₂ Tang be NEG be like Qiang.
 'Huang suspects that Tang likes Qiang.'
The 'givenness' status of the proposition: not given

$$\begin{aligned}
& \llbracket \text{是不是}_Q [\text{小唐} [\text{喜欢}] \text{小强}] \rrbracket^F \\
& = \{ \{ \lambda w. \text{Tang likes Qiang in } w, \lambda w. \text{Tang doesn't likes Qiang in } w \}, \\
& \{ \lambda w. \text{Tang hates Qiang in } w, \lambda w. \text{Tang doesn't hate Qiang in } w \}, \dots \} \\
& \Rightarrow | \llbracket \text{是不是}_Q [\text{小唐} [\text{喜欢}] \text{小强}] \rrbracket^F | > 1
\end{aligned}$$

The chosen interpretation: *huaiyi*₂

- d. 小黄 怀疑 小唐 $[\text{是不是}]$ 喜欢 小强.
‘Huang doubts (*huaiyi*₁) that Tang likes Qiang.’

The ‘givenness’ status of the proposition: given

$$\begin{aligned}
& \llbracket [\text{是不是}_Q] [\text{小唐 喜欢 小强}] \rrbracket^F \\
& = \{ \{ \lambda w. \text{Tang likes Qiang in } w, \lambda w. \text{Tang doesn't like Qiang in } w \} \} \\
& \Rightarrow | \llbracket [\text{是不是}_Q] [\text{小唐 喜欢 小强}] \rrbracket^F | = 1
\end{aligned}$$

The chosen interpretation: *huaiyi*₁

As can be seen in (25-a), when the informational focus is within the content proposition following the NQ-operator, the proposition is marked as ‘not given’, and the focus semantic value of the matrix VP is a set containing the proposition set including the highlighted (represented by the underline) content proposition and its negation, together with all other sets including alternative content propositions regarding the informational focus and their negations. In this condition, the number of members in the set denoted by the complement is more than 1, and *huaiyi*₂ is chosen for the interpretation. The other condition is triggered when the informational focus falls on the NQ-operator. In this condition the content proposition is marked as ‘given’, and the informational focus on the NQ-operator is trivial in the sense that it doesn’t introduce alternative proposition sets to the denotation of the matrix VP. Along with this, the set denoted by the complement’s focus semantic value is also singleton since no part of the proposition itself is focused and the proposition is ‘given’, with *huaiyi*₁ chosen for the interpretation of the verb predicate. The same can be observed in (25-c) and (25-d).

One thing worth noting is that the attached question ‘why was it only 11 o’clock?’ in (25-a) serves to provide a contextual clue for judging the content proposition to be ‘not given’, so that the informational focus is certain to be within the content proposition. The other condition can only be possible when this attached question is not there, as in (25-b). This proves again that both phonetic distributions and contextual information contribute to specific information structures in conformity, and their effects can both be reflected by the member number of the focus semantic value of the complement. This can also be revealed when reanalyzing (3-c) and (3-e), repeated here in (26):

- (26) a. 这 一切 老式 的 东西 让 人 怀疑
 these all old-fashioned] structural-PTCL stuff make person *huaiyi*₁
 自己 [_F是否] 还 生活 在 21 世纪.
 oneself be NEG still live in 21st century.
 ‘All these old-fashioned stuff makes one to doubt that she still lives in the 21st century.’

The ‘givenness’ status of the proposition: given

$[[[_F\text{是否}_Q][\text{自己还生活在21世纪}]]^F$

$= \{ \{ \lambda w. \text{one lives in the 21st Century in } w, \lambda w. \text{one doesn't live in the 21st Century in } w \} \}$

$\Rightarrow | [[[_F\text{是否}_Q][\text{自己还生活在21世纪}]]^F | = 1$

The chosen reading: *huaiyi*₁

- b. 许多 企业 不得不 暂时 停产, 一些
 many enterprises have to temporally stop producing, some
 外商 怀疑 上海 经济 是否 [_F过热].
 foreign entrepreneurs *huaiyi*₂ Shanghai economy be NEG overheat.
 ‘Many enterprises had to stop producing temporarily, and some foreign entrepreneurs suspected that the Shanghai economy is overheated.’

The ‘givenness’ status of the proposition: not given

$[[\text{是否}_Q[\text{上海经济}[_F\text{过热}]]]^F$

$= \{ \{ \lambda w. \text{the economy of Shanghai is overheated in } w, \lambda w. \text{the economy of Shanghai is not overheated in } w \}, \{ \lambda w. \text{the economy of Shanghai is flourishing in } w, \lambda w. \text{the economy of Shanghai is not flourishing in } w \}, \dots \}$

$\Rightarrow | [[\text{是否}_Q[\text{上海经济}[_F\text{过热}]]]^F | > 1$

The chosen reading: *huaiyi*₂

In (26-a) the contextual information of ‘the old-fashioned stuff’ serves as a ‘challenge’ against the common-sense-like ‘given’ proposition of ‘a contemporary person lives in the 21st century’, and the phonetic distribution is in conformity by putting the informational focus on the NQ-operator rather than within the proposition. In (26-b), on the other hand, the contextual information of ‘many enterprises have to temporally stop producing’ serves as the supporting evidence to bring forward the ‘not given’ proposition stating that the Shanghai economy is overheated. Accordingly, the informational focus brought by the phonetic stress falls within the proposition. If putting these observations under the light of Lu (2016), in (26-a), ‘the old-fashioned stuff’ can be regarded as an explicit immediate contextual evidence P, while the content proposition ‘she still lives in the 21st century’ is not consistent with Q in the reliable logical relationship ‘P → might Q’, for old-fashioned stuff can possibly stand for older times; whereas in (26-b), ‘many enterprises have to temporally stop producing’ is the P and the proposition ‘the Shanghai economy is overheated’ is consistent with Q.

By now the discussion has mainly focused on ‘*huaiyi*+(prototypical or quasi) B-not-B interrogative complement’ cases. Now we are turning to cases with M-not-M interrogative complements. As can be seen from examples, in contrast with the uniform B-not-B NQ operator 是不是(是否), two different forms of M-not-M NQ operator are involved in the discussion, namely 能否 (能不能) and 会不会. The former can only mean ‘can-NEG(-can)’, while the latter can mean both ‘can-NEG(-can)’ and ‘would-NEG(-would)’. The disambiguation mechanism determining the interpretation of *huaiyi* illustrated above generally still works with them, but some complications arise as can be seen in the simplified reanalysis of (3-d) and (9-b) in (27) below:

- (27) a. 人们 甚至 开始 怀疑 奥运会 [F能否] 按时 召开.
 people even start *huaiyi*₁ the Olympics can NEG on time happen.
 ‘People even started to doubt that the Olympics can happen on time.’
 $|[[[F能否]_Q][奥运会 按时 召开]]^F| = 1$
The chosen interpretation: *huaiyi*₁
- b. ?人们 甚至 开始 怀疑 奥运会 能否 [F按时 召开].
 ‘People even started to suspect (*huaiyi*₂) that the Olympics can happen on time.’
 $|[[能否]_Q[奥运会[F按时 召开]]]^F| > 1$
The chosen interpretation: *huaiyi*₂
- c. 小唐 怀疑 小邱 [F会不会] 弹 吉他.
 Tang *huaiyi*₁ Qiu can NEG can play the guitar.
 ‘Tang doubts that Qiu can play the guitar.’
 $|[[[F会不会]_Q][小邱 弹 吉他]]^F| = 1$
The chosen interpretation: *huaiyi*₁
- d. ?小唐 怀疑 小邱 会不会 [F弹 吉他].
 ‘Tang suspects (*huaiyi*₂) that Qiu can play the guitar.’
 $|[[会不会]_Q[小邱[F弹 吉他]]]^F| > 1$
The chosen interpretation: *huaiyi*₂

As seen above, it is reported that sentences with M-not-M NQ operators meaning ‘can-NEG(-can)’ have a lower acceptability when the focus semantic value of the complement is a set with more than one member, but when the focus semantic value of the complement is a singleton set, the sentences are not problematic. On the other hand, when 会不会 is interpreted as ‘would-NEG(-would)’, this problem doesn’t arise. For example, (9-b) can be analyzed alternatively as follows:

- (28) a. 小唐 怀疑 小邱 [F会不会] 弹 吉他.
 Tang *huaiyi*₁ Qiu would NEG would play the guitar.

‘Tang doubts that Qiu would play the guitar.’

$$| \llbracket [{}_F \text{会不会}_Q][\text{小邱弹吉他}] \rrbracket^F | = 1$$

The chosen interpretation: $huaiyi_1$

- b. 小唐 怀疑 小邱 会 不会 $[{}_F \text{弹吉他}]$.

‘Tang suspects ($huaiyi_2$) that Qiu would play the guitar.’

$$| \llbracket \text{会不会}_Q[\text{小邱}[_F \text{弹吉他}]] \rrbracket^F | > 1$$

The chosen interpretation: $huaiyi_2$

Therefore it can be concluded that the lower acceptability of sentences like (27-b) and (27-d) is rather peculiar and will be discussed as a remaining issue later in the last chapter. But as we’ve seen from examples above, after combining the idea of highlighting proposition in the compositional sense, the disambiguation mechanism of *huaiyi* in ‘*huaiyi*+B-not-B/M-not-M interrogative complement’ cases is similar to what have been claimed for ‘*huaiyi*+non-interrogative complement’ cases, making a distinction with the focus semantic value of the complement phrase on the formal semantic dimension: when $| \llbracket [CP]^F | > 1$, $huaiyi_2$ is chosen; while when $| \llbracket [CP]^F | = 1$, $huaiyi_1$ is chosen. With all these insights, efforts can be made to propose a unified semantics for *huaiyi*.

4.2 Approaching a unified semantics of 怀疑 (*huaiyi*)

In 3.1 we have mentioned that the semantics of *doubt* in English proposed by Anand and Hacquard (2013) can be used to account for $huaiyi_1$. $huaiyi_2$, the interpretation which is similar to *suspect* in English, can arguably be regarded as having a similar semantics with only the preference assertion (i.e. the non-equal probability judgment) reversed. Additionally, to make the semantics of *huaiyi* more unified to cover both the non-interrogative and interrogative complements, we propose to make the highlighting mechanism also relevant in the former case, in the sense that the proposition in the non-interrogative complement is also the only highlighted proposition from a singleton set of the explicit salient alternative. Based on all the investigations so far, a proposal for the semantics of *huaiyi* is given below:

- (29) For the verb *huaiyi* in the Mandarin natural language, define its metalanguage definition as BIASED-BELIEF.

1. The definition in the metalanguage:

BIASED-BELIEF(A, p, w) is defined iff

p -verifiers in $DOX_{A,w} \neq \emptyset \wedge p$ -falsifiers in $DOX_{A,w} \neq \emptyset$.

[Uncertainty Condition]

If defined = 1 iff

$$\exists w' \in DOX_{A,w} : [p(w') = 1] \wedge$$

[Doxastic Assertion]

p -falsifiers $\neq_{PROB_{A,w}}$ p -verifiers

[Preference Assertion]

where p -verifiers in $DOX_{A,w} =$

$$\lambda S'' . S'' \subset DOX_{A,w} \wedge \forall S''' \subset S'' : [\forall w' \in S''' : [p(w') = 1]] (= pow(DOX_{A,w} \cap p))$$

p -falsifiers in $DOX_{A,w} = \neg p$ -verifiers in $DOX_{A,w}$

2. The semantics of *huaiyi*:

$$\llbracket A \text{ huaiyi } \varphi \rrbracket^w = \text{BIASED-BELIEF}(A, \iota p[\text{HIGHLIGHT}(p) \wedge p \in \llbracket \varphi \rrbracket], w)$$

3. The disambiguation mechanism:

a. When φ is a clausal complement and $|\llbracket \varphi \rrbracket^F| = 1$, $\llbracket \text{huaiyi} \rrbracket = \llbracket \text{huaiyi}_1 \rrbracket$

i.e. $\exists w' \in DOX_{A,w} : [p(w') = 1] \wedge p$ -falsifiers $>_{PROB_{A,w}}$ p -verifiers, where $p = \iota p[\text{HIGHLIGHT}(p) \wedge p \in \llbracket \varphi \rrbracket]$.

b. When φ is a clausal complement and $|\llbracket \varphi \rrbracket^F| > 1$, $\llbracket \text{huaiyi} \rrbracket = \llbracket \text{huaiyi}_2 \rrbracket$

i.e. $\exists w' \in DOX_{A,w} : [p(w') = 1] \wedge p$ -verifiers $>_{PROB_{A,w}}$ p -falsifiers, where $p = \iota p[\text{HIGHLIGHT}(p) \wedge p \in \llbracket \varphi \rrbracket]$.

From the first sight, the accounts given above are similar to the semantics of *doubt* in Anand and Hacquard (2013). But one should notice that in Anand and Hacquard (2013), the preference assertion gives a settled unidirectional relationship between the probability of p -falsifiers and p -verifiers in the subject's doxastics within the evaluation world, namely p -falsifiers are more probable than p -verifiers. Whereas for the Mandarin *huaiyi*, due to its characteristics, the preference assertion only manifests a bias, or an inequality, between the probability of p -falsifiers and p -verifiers for the subject in the evaluation world. This inequality can be bidirectional, and the specific direction is determined by the disambiguation mechanism.

Besides, it's worth noting that although the disambiguation mechanism proposed above is generally applicable in the empirical sense, it's still stipulative, requiring at least an attempt to account for the reason behind it. We hold that this mechanism can be regarded as a manifestation of the fundamental disambiguation criterion of logical relationship claimed by Lu (2016), supported by the conformity between phonetic distributions and contextual clues in the interpretation observed in 4.1.2. More specifically, the immediate contextual evidence 'P' is always relevant in the interpretation of *huaiyi*, either explicitly (as contextual clues in sentences of (25-b) and (26)) or implicitly. When it is not directly accessible, the proposed disambiguation mechanism and the information structure behind it can serve as the clue to bring it forward. Empirically, when the content proposition in the complement is consistent with the reasoning result of P (i.e. Q as in the $P \rightarrow \text{might } Q$), the informational focus tends to

lie in the proposition, whereas when the content proposition is in conflict with the reasoning result from P (put under the label of $\neg Q$), the informational focus tends to be out of it, falling on *huaiyi* itself in non-interrogative complements and the NQ-operator in interrogative complements. For instance, as for sentences from (4), before their utterances, there would definitely be an implicit observation about the weather clues concerning the possibility of a sunny weather during the next day, which can be regarded as an immediate contextual evidence ‘P’. When the content proposition 明天会晴 (‘it will be sunny tomorrow’) is consistent with this implicit ‘P’, the phonetic distribution and the corresponding information structure are realized as in (4-a) and (4-b); when it is not consistent with ‘P’, the sentence is realized as (4-c). One possible reason behind this is that the logical reasoning behind the dubitative attitude is an immediate process, if the agent draws a proposition that is consistent with the latest immediate contextual evidence, it tends to be new, prominent and ‘not given’, leading to ‘ $|\llbracket \varphi \rrbracket^F| > 1$ ’ on the formal dimension, and the attitude is the positive *huaiyi*₂. In contrast, when the proposition is not consistent with the latest immediate contextual evidence, the necessity of bringing it into the discourse tends to originate from its past existence and ‘givenness’, leading to ‘ $|\llbracket \varphi \rrbracket^F| = 1$ ’ on the formal dimension, and the attitude is the negative *huaiyi*₁.

5 Conclusions and remaining issues

By now we have attempted to resolve the two main research questions stated in the introduction: 1) What is the special disambiguation mechanism at work for the Mandarin dubitative verb *huaiyi*? 2) How to construct a relatively unified semantics of *huaiyi* in a formal framework with the combination of the aforementioned mechanism and the two empirical interpretations *huaiyi*₁ and *huaiyi*₂? For the first question, we have concluded that there is a widely-applicable disambiguation mechanism to distinguish the two interpretations of *huaiyi*₁ and *huaiyi*₂ based on the information structure, mainly depending on the location of the informational focus and the ‘givenness’ status of the content proposition in the embedding complement. On the dimension of the formal expression, this mechanism is revealed in the value of $|\llbracket \varphi \rrbracket^F|$. When the informational focus is outside the content proposition and the proposition is marked as ‘given’, $|\llbracket \varphi \rrbracket^F| = 1$ and *huaiyi*₁ is chosen for the interpretation; when the informational focus is within the content proposition and the proposition is marked as ‘not given’, $|\llbracket \varphi \rrbracket^F| > 1$ and *huaiyi*₂ is chosen for the interpretation. It has also been claimed that this mechanism is derived from the logical relationship between the immediate contextual evidence and the content proposition: the former is the case when the proposition is not consistent with the reasoning from the immediate contextual evidence,

while the latter is the case when the proposition is consistent with the reasoning from the immediate contextual evidence.

As for the second question, having referred to the research on the English dubitative verb *doubt*, we claim that *huaiyi*, as a dubitative verb in Mandarin Chinese, also possesses a hybrid semantics. Specifically, it existentially quantifies over one's doxastic state, conveying that some worlds in the doxastic state verifies the content proposition, but also allows some worlds in which the proposition is falsified. Meanwhile, there is also a preference assertion regarding the probability, conveying that for the agent, there is a bias such that the probability of being compatible with the real world regarding verifying worlds and falsifying worlds are not equal. The direction towards which this bias leans is determined by the disambiguation mechanism that has been discussed.

It's clear that there are still unsolved remaining issues in this thesis, and two of them will be discussed here. Firstly, in our proposed disambiguation mechanism for *huaiyi*, nominal complements are not included. The reason is that as mentioned in the section 2.1, when embedding a nominal complement, *huaiyi* seems to be interpreted undisputedly as *huaiyi*₁ without ambiguity, and the alternations in the information structure brought by the distribution of informational focus we've been discussing for clausal complements seems to be irrelevant in such cases. Here a premature speculation is that the grammatical definiteness of a nominal complement has decisively marked itself as 'given' in the information structure, which forces the interpretation towards *huaiyi*₁ and creates a blocking effect against other mechanisms. Secondly, as observed in (27-b) and (27-d) of the section 4.1.2, there is a lower acceptability for sentences with M-not-M operators denoting 'can-NEG(-can)' when the focus semantic value of the complement is a set with more than one member. This is not accounted for in our proposal, and a tendency to explain this is to attribute it as a pragmatic problem where the idea of X. Li (1987) about 'interrogatives serving as euphemism' is at work: since M-not-M operators denoting 'can-NEG(-can)' are very likely to scope over propositions related to 'desirable' abilities, when the intended message is *huaiyi*₂ + something 'desirable', there is no need for the 'euphemism', namely using the interrogative structure to bring forward a 'undesirable' proposition indirectly. People may prefer to use '*huaiyi* + non-interrogative complement' directly in such cases. This may have led to the peculiarity for sentences like (27-b) and (27-d), although the proposed disambiguation mechanism can work in them principally. All in all, the solution to these two remaining issues may require more research into the topic of this thesis.

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